# Vocabulary: Explainable AI – Week 02
# Global Explanations: Feature Effects

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## Global Explanations

- **Global Explanation**: A description of a model's overall behavior across a dataset, rather than an explanation for one individual prediction. PFI, PDP, ICE, and ALE are examples introduced in the lecture.
- **Feature Importance**: A measure of how much a feature matters to a model's predictions or performance. The lecture uses Permutation Feature Importance to answer "How much does this feature matter?"
- **Feature Effect**: A description of how predictions change as a feature changes. The lecture uses PDP, ICE, and ALE to answer "How does this feature matter?"
- **Model-agnostic Method**: An explanation method that only needs the model's inputs and outputs, such as predictions from `model.predict`. PFI, PDP, ICE, and ALE are all model-agnostic.
- **Post-hoc Explanation**: An explanation computed after a model has been trained. The global methods in this lecture are post-hoc because they analyze an already-trained model.
- **Descriptive Explanation**: An explanation of what a model has learned, not proof of a real-world causal relationship. The lecture emphasizes that PFI, PDP, ICE, and ALE describe the model rather than establish causality.

## Permutation Feature Importance

- **Permutation Feature Importance (PFI)**: A global method that measures importance by shuffling one feature and observing how much model performance worsens. Formally, it can be written as:

$$\text{PFI}_j = s - s_{\text{perm},j}$$

  where $s$ is the original score and $s_{\text{perm},j}$ is the score after permuting feature $j$.
- **Baseline Score**: The model's performance on unmodified held-out data before any feature is permuted. PFI compares the score after permutation with this baseline.
- **Permutation**: Randomly reordering the values of one feature column while keeping the other columns unchanged. This destroys the relationship between that feature and the target.

- **Held-out Data**: Data not used to train the model, typically a test set. The lecture recommends computing PFI on held-out data because an overfit model can assign importance to noise features on training data.
- **Repeated Permutation**: Running the shuffle-and-score procedure multiple times. The lecture recommends reporting both the mean importance and its spread because permutation introduces randomness.
- **Correlated Features**: Features whose values tend to vary together. With PFI, shuffling one correlated feature can create unrealistic rows and can split importance between the correlated features.
- **Grouped Permutation**: Permuting a group of related features together rather than one at a time. The lecture presents this as a way to reduce problems caused by correlated features.
- **Importance versus Direction**: PFI indicates how much a feature affects performance, but not whether increasing the feature raises or lowers the prediction. Effect plots such as PDP, ICE, and ALE address direction and shape.

## PDP and ICE

- **Partial Dependence Plot (PDP)**: A global effect plot showing the average model prediction for different values of one feature, averaging over the observed values of all other features:

$$\hat{f}_S(x_S) = \mathbb{E}_{X_C}\left[f(x_S, X_C)\right] \approx \frac{1}{n}\sum_{i=1}^{n} f\left(x_S, x_C^{(i)}\right)$$

- **Feature Grid**: A set of values for the feature of interest. A PDP requires predicting at every grid value for every instance, then averaging the predictions at each grid value.
- **Marginal Averaging**: Averaging over the observed distribution of the other features. In a standard PDP, this can create feature combinations that do not occur in the real data.
- **Independence Assumption**: The PDP assumption that the feature of interest is independent of the remaining features. The lecture warns that this assumption is problematic when features are correlated.
- **Rug Plot**: Small marks along the axis of a plot showing where observed data values occur. The lecture recommends adding a rug or histogram to a PDP to see where the curve is supported by data.
- **Extrapolation**: Evaluating a model in regions with little or no training data. PDPs can extrapolate when they average over unrealistic combinations of correlated features.
- **Individual Conditional Expectation (ICE)**: A plot showing one prediction curve per instance as the feature of interest varies, while that instance's other feature values remain fixed:

$$\text{ICE}^{(i)}(x_S) = f\left(x_S, x_C^{(i)}\right)$$

- **Heterogeneity**: Differences in how the feature affects predictions for different instances. ICE curves reveal heterogeneity that a PDP may hide by averaging.

- **PDP as Average ICE**: The PDP is the pointwise average of the ICE curves. If ICE curves cross or point in opposite directions, their average may look flat or misleading.
- **Centered ICE Plot**: An ICE variant that shifts each individual curve to a common starting point, making differences in curve shape easier to compare.

## Accumulated Local Effects

- **Accumulated Local Effects (ALE)**: A global effect method designed to handle correlated features by computing local prediction differences within intervals of observed data and accumulating them:

$$\hat{f}_{j,\mathrm{ALE}}(x) = \sum_{k=1}^{k_j(x)} \frac{1}{n_k} \sum_{i:\, x_j^{(i)} \in N_k} \left[ f\left(z_k, x_{\setminus j}^{(i)}\right) - f\left(z_{k-1}, x_{\setminus j}^{(i)}\right) \right] - c$$

- **Quantile Bins**: Intervals created from quantiles of the feature of interest. ALE divides the feature range into bins so that each bin contains observed data.
- **Local Prediction Difference**: The change in prediction when a feature is moved from the lower to the upper edge of its bin, while the instance's other features remain unchanged.
- **Accumulation**: Adding the local effects across bins. This produces an overall effect curve for the feature.
- **Centering Constant**: A constant $c$ that shifts the ALE curve so that its average effect is zero. This makes the curve easier to interpret relative to the mean prediction.
- **ALE Interpretation**: The ALE value at a feature value describes the average local change in prediction relative to the mean prediction, rather than an absolute predicted level.
- **ALE versus PDP**: PDP estimates the average prediction at a feature value, while ALE estimates the average local change around that value. ALE is designed to be more robust when features are correlated.

## Interactions & Method Choice

- **Feature Interaction**: A situation in which the effect of one feature depends on the value of another feature. For example, the effect of latitude on housing price may depend on longitude.
- **Main Effect**: The effect of one feature on the prediction when considered separately from other features. One-dimensional PDPs and ALE plots show main effects.
- **Two-dimensional PDP**: A PDP that shows the joint effect of two features, such as latitude and longitude. The lecture uses it to detect interaction effects that one-dimensional plots can miss.
- **Friedman's $H$-statistic**: A measure of the share of prediction variance attributable to interactions. The lecture lists it as a way to detect feature interactions.
- **Model Behavior versus Causal Truth**: The distinction between explaining what a trained model predicts and explaining what actually causes an outcome. Global explanation plots describe the former, not the latter.

## Common XAI Shortnames

| **Shortname** | **Meaning** |
|---|---|
| PFI | Permutation Feature Importance; measures the performance drop after shuffling a feature |
| PDP | Partial Dependence Plot; shows the average prediction effect of a feature |
| ICE | Individual Conditional Expectation; shows one effect curve per instance |
| ALE | Accumulated Local Effects; computes local effects within observed data bins |
| 2D PDP | Two-dimensional Partial Dependence Plot; shows the joint effect of two features |
| $H$ | Friedman's $H$-statistic; measures the strength of feature interactions |
| XAI | Explainable Artificial Intelligence |